

Survey Research and Design in Psychology

Tutorial 3 - Psychometrics

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- ▶ Talk about the general steps for psychometric instrument development
- ▶ Exploratory factor analysis
- ▶ Internal consistency
- ▶ Creating composite scores
- ▶ Sample write ups

Steps

- 1) Check and prepare the data
- 2) Exploratory factor analysis
- 3) Analyse internal consistency for each factor
- 4) Examine correlations amongst the factors
- 5) Compute composite scores if needed
- 6) Check factor structure across subgroups

Check and prepare data

- ▶ Clean the data (for lab report, follow **data screening instructions**)
- ▶ Reverse code variables where needed so they are all scored in the same direction - necessary for reliability
- ▶ Examine the univariate properties (e.g. histogram, mean, standard deviations, skewness, kurtosis)
- ▶ Examine the bivariate properties (e.g. scatterplots and correlations)

- ▶ Write out your research question (e.g. what are the underlying dimensions of X?)
- ▶ Check the assumptions (sample size, linearity, homoscedasticity, factorability)
- ▶ Choose your method of extraction (principal components analysis/principal axis factoring) and rotation (orthogonal/varimax or oblique/oblimin) and conduct the exploratory factor analysis
- ▶ Determine the number of factors
- ▶ Name and describe each factor
- ▶ Identify the items that belong to each factor, remove problematic items and repeat analysis

- ▶ Can we summarise attitudes towards other people smoking in a few factors?
- ▶ On a scale from 1 (strongly disagree) to 100 (strongly agree), please indicate the extent to which you agree with each of the following statements:
 - ▶ I think smoking is acceptable (question 1)
 - ▶ I don't care if people smoke around me (question 2)
 - ▶ I don't think people should smoke in restaurants (question 3)
 - ▶ I think people should have the right to smoke (question 4)
 - ▶ I don't think people should smoke around food (question 5)

Download Tutorial 3 - dataset 1

Tutorial 3 - dataset 1

- 1) Check and prepare the data - for today, let's assume this step has been done already
- 2) Exploratory factor analysis
- 3) Analyse internal consistency for each factor
- 4) Examine correlations amongst the factors
- 5) Compute composite scores if needed
- 6) Check factor structure across subgroups

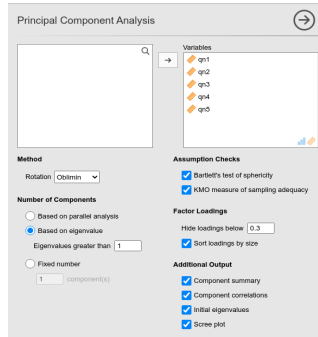
- ▶ Theory - question is about reducing items into smaller number of factors, so we will use principal components analysis
- ▶ Check the following assumptions:
 - ▶ Sample size - what is the cases to variables ratio?
 - ▶ At a minimum: more than 5 cases per variable (1:5 variables:cases ratio)
 - ▶ Ideal: more than 20 cases per variable (1:20 variables:cases ratio)
 - ▶ Level of measurement - what is the level of measurement of your data
 - ▶ interval or ratio level of measurement
 - ▶ Normality - check the histograms
 - ▶ Factor analysis can cope with minor violations of normality (i.e. it is relatively robust). However, it works best if the variables are normally distributed

- ▶ Next assumptions to check - are variables correlated? In other words, are your data factorable? Get a correlation matrix between all 5 variables
 - ▶ Correlation matrix correlations $> .3$
- ▶ Analyses $>$ Regression $>$ correlation matrix $>$ move across all variables (qn1 - qn5)

- ▶ Are the associations between variables best explained by a linear correlation
- ▶ Plot
 - ▶ Correlation matrix

Assumptions

- ▶ Run an initial factor analysis and test the remaining assumptions
- ▶ Analyses > Factor > Principal Components Analysis > move across all variables (qn1 - qn5)



The screenshot shows the 'Principal Component Analysis' dialog box in SPSS. The 'Variables' list on the right contains qn1, qn2, qn3, qn4, and qn5. Under 'Method', the 'Rotation' is set to 'Oblimin'. Under 'Number of Components', 'Based on eigenvalue' is selected with a threshold of 1. Under 'Assumption Checks', both 'Bartlett's test of sphericity' and 'KMO measure of sampling adequacy' are checked. Under 'Factor Loadings', 'Hide loadings below' is set to 0.3 and 'Sort loadings by size' is checked. Under 'Additional Output', 'Component summary', 'Component correlations', 'Initial eigenvalues', and 'Scree plot' are all checked.

Principal Component Analysis

Variables

- qn1
- qn2
- qn3
- qn4
- qn5

Method

Rotation: Oblimin

Number of Components

- ☐ Based on parallel analysis
- ☒ Based on eigenvalue
- Eigenvalues greater than: 1
- ☐ Fixed number
- 1 component(s)

Assumption Checks

- ☒ Bartlett's test of sphericity
- ☒ KMO measure of sampling adequacy

Factor Loadings

Hide loadings below: 0.3

- ☒ Sort loadings by size

Additional Output

- ☒ Component summary
- ☒ Component correlations
- ☒ Initial eigenvalues
- ☒ Scree plot

- ▶ Bartlett's significance?
- ▶ $KMO > .5$ or $.6$?

How many factors?

- ▶ To work this out, you can check a number of places:
 - ▶ Scree plot
 - ▶ Percentage of variance explained
 - ▶ If the solution is rotated (i.e. either an orthogonal or oblique rotation is used) then the variance from the component statistics table should be used. If unrotated, then either the components statistics table or eigenvalues table should be used, as results will be the same. Eigenvalues table does not adjust variance explained with rotation method.
 - ▶ Factor loadings and meanings of factors

Two factor solution

- ▶ Factor 1
 - ▶ I think people should have the right to smoke (question 4)
 - ▶ I think smoking is acceptable (question 1)
 - ▶ I don't care if people smoke around me (question 2)
- ▶ Factor 2
 - ▶ I don't think people should smoke around food (question 5)
 - ▶ I don't think people should smoke in restaurants (question 3)

Orthogonal vs oblique - reminder

- ▶ Are you expecting factors to be related or unrelated?
 - ▶ If related, choose oblique (oblimin in jamovi)
 - ▶ If unrelated, choose orthogonal (varimax in jamovi)
 - ▶ If unsure, choose oblique, then switch to orthogonal if the correlations between factors are smaller than .3
 - ▶ You can try both, then see which solution is most interpretable
- ▶ For this question, the factors are interpretable, but there is only a weak correlation between factor 1 and 2
 - ▶ Change to a varimax rotation

Write up - things to note

- ▶ Type of factor analysis?
- ▶ Rotated or not? If so, what kind?
- ▶ Did you meet assumptions?
- ▶ How many factors were extracted?
- ▶ What percentage of variance did these explain?
- ▶ Interpret - what were the factors

Question

- ▶ Teachers rated students across 15 classroom behaviours
- ▶ Can we summarise classroom behaviours into a smaller number of factors?

Download Tutorial 3 - dataset 2

Tutorial 3 - dataset 2

1	Concentrates	Cannot concentrate on any particular task; easily distracted ↔ can concentrate on any task; not easily distracted
2	Curious	Shows little curiosity or motivation ↔ eager to learn; curious and enquiring
3	Perseveres	Lacks perseverance; is impatient with difficult or challenging work ↔ perseveres in the face of difficult or challenging work
4	Even-tempered	Irritable, 'touchy', 'cranky' ↔ even-tempered
5	Placid	Easily excited; gets 'high' ↔ not easily excited; placid
6	Compliant	Demanding and argumentative ↔ patient and compliant
7	Self-controlled	Has difficulty controlling own behaviour ↔ is able to control own behaviour
8	Relates-warmly	Provocative; disruptive; short attention span ↔ relates warmly to others
9	Sustained attention	Easily frustrated; short attention span ↔ persistent, sustained attention span
10	Communicative	Difficult to reason and communicate with ↔ easy to reason and communicate with
11	Relaxed	Restless; fidgety; can't sit still ↔ relaxed; can sit still
12	Calm	On the go; lively; always moving ↔ settled, calm
13	Purposeful activity	Aimless activity ↔ purposeful activity
14	Cooperative	Disputes, fights over sharing ↔ co-operative; shared with others
15	Contented	Is easily upset; unhappy ↔ contented; happy

- 1) Check and prepare the data
- 2) Exploratory factor analysis
- 3) Analyse internal consistency for each factor
- 4) Examine correlations amongst the factors
- 5) Compute composite scores if needed
- 6) Check factor structure across subgroups

- ▶ Start with a correlation matrix - are your data factorable?
- ▶ Analyses > Regression > correlation matrix > move across all variables (behav1 - behav15)

Assumptions

- ▶ Run an initial factor analysis and test the remaining assumptions
- ▶ Analyses > Factor > Principal Components Analysis > move across all variables (behav1 - behav15)

The screenshot shows the 'Principal Component Analysis' dialog box in SPSS. The 'Variables' list on the right contains 'behav1', 'behav2', 'behav3', 'behav4', 'behav5', 'behav9', and 'behav10'. Under 'Method', 'Rotation' is set to 'Oblimin'. Under 'Number of Components', 'Based on eigenvalue' is selected with a threshold of 1. Under 'Assumption Checks', both 'Bartlett's test of sphericity' and 'KMO measure of sampling adequacy' are checked. Under 'Factor Loadings', 'Hide loadings below' is set to 0.3 and 'Sort loadings by size' is checked. Under 'Additional Output', 'Component summary', 'Component correlations', 'Initial eigenvalues', and 'Scree plot' are all checked.

Principal Component Analysis

Variables

- behav1
- behav2
- behav3
- behav4
- behav5
- behav9
- behav10

Method

Rotation: Oblimin

Number of Components

- ☐ Based on parallel analysis
- ☒ Based on eigenvalue
- Eigenvalues greater than: 1
- ☐ Fixed number
- 2 component(s)

Assumption Checks

- ☒ Bartlett's test of sphericity
- ☒ KMO measure of sampling adequacy

Factor Loadings

Hide loadings below: 0.3

- ☒ Sort loadings by size

Additional Output

- ☒ Component summary
- ☒ Component correlations
- ☒ Initial eigenvalues
- ☒ Scree plot

- ▶ Bartlett's significance?
- ▶ $KMO > .5$ or $.6$?

How many factors?

- ▶ To work this out, you can check a number of places:
 - ▶ Scree plot
 - ▶ Percentage of variance explained
 - ▶ Factor loadings and meanings of factors

How many factors

- ▶ You can change the number of factors extracted
 - ▶ Fixed number > 3 components

Factor 1

- ▶ Perseveres - behav3
- ▶ Curious - behav2
- ▶ Purposeful activity - behav13
- ▶ Concentrates - behav1
- ▶ Sustained attention - behav9

Factor 2

- ▶ Relates warmly - behav8
- ▶ Contented - behav15
- ▶ Cooperative - behav14
- ▶ Even-tempered - behav4
- ▶ Communicative - behav10

Factor 3

- ▶ Placid - behav5
- ▶ Calm - behav12
- ▶ Relaxed - behav11
- ▶ Compliant - behav6
- ▶ Self-controlled - behav7

Selecting items depends on...

- ▶ An exploratory factor analysis of a good measurement instrument ideally has:
 - ▶ A simple factor structure (each variable loads strongly ($> .50$) on only one factor)
 - ▶ Each factor has multiple loading variables (more loadings $\rightarrow$ greater reliability)
 - ▶ Target factor loadings are high ($> .5$) and cross-loadings are low ($< .3$), with few intermediate values (.3 to .5)
- ▶ We could consider removing `behav6` and `7` and re-examine the factor structure

Factor 1

- ▶ Perseveres - behav3
- ▶ Curious - behav2
- ▶ Purposeful activity - behav13
- ▶ Concentrates - behav1
- ▶ Sustained attention - behav9

Factor 2

- ▶ Relates warmly - behav8
- ▶ Contented - behav15
- ▶ Cooperative - behav14
- ▶ Even-tempered - behav4
- ▶ Communicative - behav10

Factor 3

- ▶ Placid - behav5
- ▶ Calm - behav12
- ▶ Relaxed - behav11

Write up - things to note

- ▶ Type of factor analysis?
- ▶ Rotated or not? If so, what kind?
- ▶ Did you meet assumptions?
- ▶ How many factors were extracted?
- ▶ What percentage of variance did these explain?
- ▶ Interpret - what were the factors

- ▶ Are there underlying dimensions in managerial skills?

Download Tutorial 3 - dataset 3

Tutorial 3 - dataset 3

To what extent do you possess these skills

- 1) I show confidence in my staff
- 2) I let my staff know when they are doing well
- 3) I give feedback to staff on how well they are working
- 4) I would personally compliment staff if they did outstanding work
- 5) I believe in setting goals and achieving them
- 6) I achieve the things I want to get done in a day
- 7) I never try to put off until tomorrow what I can finish today
- 8) I plan the use of my time well
- 9) I remain clear headed when too many demands are made upon me
- 10) I rarely overlook important factors when plans are made
- 11) I handle complex problems efficiently

Steps - have a go!

- 1) Check correlation matrix
- 2) Principal component analysis with oblimin rotation
- 3) KMO/Bartlett's
- 4) Variance explained, examine eigenvalues and scree plots
- 5) Component loading matrix

► How many factors are there? What are they?

Question

- ▶ Do the items that measure openness form a reliable scale

Download Tutorial 3 - dataset 4

Tutorial 3 - dataset 4

- 1) Believe in the importance of art
- 2) Tend to vote for conservative political candidates
- 3) Have a vivid imagination
- 4) Avoid philosophical discussions
- 5) Tend to vote for liberal political candidates
- 6) Do not like art
- 7) Carry the conversation to a higher level
- 8) Am not interested in abstract ideas
- 9) Enjoy hearing new ideas
- 10) Do not enjoy museums

- ▶ For a multi item scale, how do we know if the items work well together to form a reliable scale?

Step 1 - Reverse coding

- ▶ Items 2, 4, 6, 8 and 10 are measured in the opposite direction of the remaining items. For example:
 - ▶ Item 1 - “Believe in the importance of art”, is a pro-trait. The higher you score, the more open you are to experience
 - ▶ Item 6 - “Do not like art” is a con-trait. The lower you score the more open you are to experience
- ▶ We need to recode con-trait items so that all items point in the same direction i.e. higher scores = greater openness

- ▶ Right click variable 2 > Transform > using transform > create new transformation
- ▶ Name - "Reverse code", Description - "Reverse codes a 1 to 5 scale, Suffix - R
- ▶ Type, 6 - \$source

Recoding variables

- ▶ Scale was 1 to 5
- ▶ Formula is $6 - \text{original value}$ (in jamovi, `$source`)
- ▶ How does that work?
 - ▶ $1 = 5$
 - ▶ $2 = 4$
 - ▶ $3 = 3$
 - ▶ $4 = 2$
 - ▶ $5 = 1$

- ▶ Now that we have recoded Open, we want to apply the same transformation to the other variables that need it (Open 4, 6, 8 and 10)
- ▶ Right click variable 2 > Transform > using transform > “Reverse code”

- ▶ Now that all variables are 'pointing' in the same direction, we can see how well they fit together to form a reliable scale
- ▶ Analyses > Factor > Reliability Analysis
- ▶ Note: Be careful to add all 'unrecoded' odd number items and all recoded even number items. In other words, ensure the even numbered items that do not end in -R stay on the left side of your window

- ▶ Scale statistic
 - ▶ Cronbach's α
- ▶ Item statistics
 - ▶ Cronbach's α (if item is dropped)
 - ▶ Item-rest correlation
- ▶ Is the scale reliable overall?
 - ▶ $< .6$ = Unreliable
 - ▶ $.6$ = OK
 - ▶ $.7$ = Good
 - ▶ $.8$ = Very good, strong
 - ▶ $.9$ = Excellent
 - ▶ $> .95$ = may be overly reliable or redundant - depends on what is being measured

- ▶ Looking at the item reliability statistics, are there any items that do not appear to fit with the rest?
 - ▶ Open 5?
- ▶ Open 5: Tend to vote for liberal political candidates.
 - ▶ This item was not related in the expected direction, and removing the item would improve the reliability of the scale
 - ▶ Perhaps because the meaning of 'liberal' is different in Australia, or confused with the Liberal Party
 - ▶ Take Open5 out of the analysis and check again

- ▶ We now know that we have a reliable scale to measure Openness with the following items
 - ▶ Open1, Open2 - R, Open3, Open4 - R, Open6 - R, Open7, Open 8 - R, Open 9, Open 10 - R
- ▶ Data > Compute
 - ▶ Give the mean score a name (i.e. OpenM) and a descriptor (i.e. Mean openness score)
 - ▶ Click f_x to change the function to a MEAN and you will see instructions on how to create this - note that you put commas between variable names.
 - ▶ Double clicking the MEAN function will bring this into the formula box, and then within the brackets, double click to select variables and enter the commas yourself.

Create a mean score

- ▶ You can now check the descriptive stats of the variable you have just created (i.e. a composite openness score - higher scores = greater openness)
- ▶ Analyses > Exploration > Descriptives > Move OpenM into the variables box
 - ▶ Plots
 - ▶ Histogram

Questions?

- ▶ You can ask questions any time on the discussion forums.
 - ▶ Jamovi
 - ▶ Tutorial